

SHENANDOAH, IA, USA

WMO: 725467

Lat: 40.751N

Lon: 95.413W

Elev: 971

StdP: 14.19

Time Zone: -6.00 (NAC)

Period: 95-19

WBAN: 04973

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-2.8	2.7	-9.4	3.4	-0.1	-6.0	4.1	4.7	26.5	32.7	24.3	31.2	4.2	0	0.526

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.3	95.3	78.9	92.7	77.4	90.3	76.5	82.3	91.7	80.2	89.7	78.3	87.3	11.4	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
80.5	164.3	89.9	77.3	147.2	86.7	75.3	137.4	84.2	46.9	92.6	44.7	90.0	42.4	87.1	89.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
24.9	21.4	18.9	-13.9	97.8	7.1	3.2	-19.0	100.1	-23.1	102.0	-27.1	103.8	-32.2	106.1		
			-14.9	85.0	6.6	2.4	-19.7	86.7	-23.5	88.1	-27.2	89.4	-32.0	91.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	52.7	25.8	29.2	41.3	52.8	64.3	73.5	77.2	74.3	67.2	54.9	40.8	29.7
	DBStd	20.54	12.47	12.65	12.20	10.09	8.58	6.50	6.23	5.80	8.48	9.53	10.29	11.22
	HDD50	2742	749	586	321	83	5	0	0	0	1	57	305	635
	HDD65	5699	1214	1002	736	382	119	9	2	4	75	334	727	1095
	CDD50	3737	0	4	52	168	449	706	843	752	517	211	30	5
	CDD65	1220	0	0	3	17	98	265	381	292	141	22	1	0
	CDH74	12865	0	0	37	248	1002	2762	4217	2794	1485	291	13	16
	CDH80	5314	0	0	4	74	360	1148	1935	1123	574	84	1	11
Wind	WSAvg	8.0	8.2	8.6	10.0	10.6	9.5	7.9	5.7	5.2	6.5	7.7	8.3	7.8
Precipitation	PrecAvg	34.9	0.8	1.1	1.9	3.1	5.6	5.8	4.2	3.5	2.8	2.3	1.6	1.0
	PrecMax	52.6	2.0	3.2	4.4	8.2	11.6	14.7	14.3	9.3	7.0	8.3	4.4	3.1
	PrecMin	26.1	0.1	0.1	0.0	0.8	1.6	2.2	0.0	1.0	0.4	0.0	0.0	0.0
	PrecStd	6.7	0.5	0.7	1.3	1.6	2.8	2.8	3.1	2.1	1.7	2.0	1.3	0.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	60.7	67.3	78.9	89.7	94.8	97.0	100.0	97.0	95.1	88.2	75.2	65.5
		MCWB	46.5	54.0	59.9	66.3	73.4	78.0	80.5	79.9	75.3	69.2	59.9	60.4
	2%	DB	53.6	59.3	72.5	81.6	88.4	93.0	96.8	92.8	90.6	81.5	68.3	56.5
		MCWB	44.7	49.7	58.0	62.8	70.5	75.9	81.4	78.7	73.8	64.1	56.4	49.1
	5%	DB	46.6	53.7	66.2	75.5	83.7	90.3	92.8	90.1	86.4	76.7	63.0	51.6
		MCWB	40.7	45.6	55.1	60.3	68.1	75.4	80.0	77.9	72.3	62.4	52.7	45.8
	10%	DB	42.7	46.2	61.3	71.6	80.7	86.8	90.2	86.3	82.2	72.1	57.1	45.6
		MCWB	37.7	40.8	52.4	58.2	66.3	73.3	78.7	76.3	69.4	60.0	48.5	40.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	49.4	56.8	64.3	69.7	77.6	82.3	85.9	83.7	78.3	73.2	62.5	60.0
		MCDB	55.3	62.6	73.1	80.1	89.9	91.4	95.9	93.2	90.6	82.6	71.5	64.6
	2%	WB	45.1	50.5	60.6	65.8	73.7	79.5	83.3	81.2	76.3	68.4	58.2	50.8
		MCDB	51.3	59.4	68.0	77.3	82.2	89.5	92.7	89.6	87.0	75.6	65.5	53.8
	5%	WB	41.4	45.5	57.5	62.4	70.7	77.2	81.2	79.1	74.3	65.4	53.9	44.9
		MCDB	47.3	52.5	65.3	73.0	80.3	86.9	90.9	87.7	83.8	72.6	61.6	50.1
	10%	WB	37.7	41.2	52.6	59.4	68.1	75.1	79.3	77.2	72.2	61.5	49.9	40.7
		MCDB	42.2	46.6	61.8	69.2	76.9	84.4	88.7	85.2	80.2	70.7	57.1	45.8
Mean Daily Temperature Range	5% DB	MDBR	19.1	19.3	21.7	23.5	21.0	20.4	20.3	20.6	24.4	24.2	21.6	18.6
		MCDBR	26.9	28.1	31.8	31.5	26.6	22.9	22.7	23.6	27.0	30.5	29.9	26.3
		MCWBR	19.6	19.0	18.8	16.7	12.1	10.6	12.1	12.9	12.9	16.1	18.6	19.2
	5% WB	MCDBR	25.5	26.2	25.7	25.2	21.7	20.6	20.5	21.3	23.3	23.2	26.9	23.5
		MCWBR	19.1	18.7	16.7	15.4	11.9	11.3	12.3	13.1	12.4	14.8	18.5	18.3
Clear-Sky Solar Irradiance	taub	0.280	0.315	0.326	0.382	0.410	0.416	0.427	0.420	0.377	0.334	0.297	0.285	
	taud	2.468	2.380	2.415	2.278	2.268	2.300	2.260	2.297	2.370	2.478	2.511	2.464	
	Ebn at Noon	282	285	294	282	275	273	269	267	272	275	273	270	
	Edh at Noon	24	31	33	41	42	41	43	40	35	28	23	23	
All-Sky Solar Radiation	RadAvg	657	943	1226	1550	1840	2043	2051	1790	1485	1039	723	552	
	RadStd	50	62	112	98	151	162	143	100	98	120	56	56	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (42 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page